

SONORA, TX, USA

WMO: 747335

Lat: 30.586N

Lon: 100.649W

Elev: 2140

StdP: 13.59

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03059

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.3	26.6	7.3	8.7	37.2	10.2	10.0	40.3	22.3	47.8	19.9	50.8	2.7	20	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.5	99.4	69.6	97.4	69.4	95.4	69.6	75.2	85.7	74.3	84.6	73.5	83.6	9.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	130.6	77.9	72.2	129.0	77.7	71.7	126.7	77.4	40.1	86.0	39.3	84.5	38.6	83.0	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.9	18.1	16.5	DB	15.4	102.5	2.8	2.9	13.3	104.6	11.7	106.2	10.2	107.8	8.1	109.9	(4)
			WB	13.7	76.6	2.8	1.0	11.7	77.3	10.1	77.9	8.5	78.4	6.5	79.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	66.2	46.6	51.7	59.9	67.0	74.0	81.5	82.8	83.0	76.3	66.9	55.6	47.7	(6)
		DBStd	14.98	8.14	9.70	9.04	7.72	6.90	4.38	3.85	4.10	5.47	8.64	8.70	8.09	(7)
		HDD50	470	162	88	25	2	0	0	0	0	5	44	144		(8)
		HDD65	2155	572	377	202	68	12	0	0	2	85	299	538		(9)
		CDD50	6365	56	136	333	513	745	944	1016	1022	790	528	211	71	(10)
		CDD65	2575	1	5	44	129	292	494	551	557	343	143	16	0	(11)
		CDH74	30256	37	157	509	1410	3139	6020	6798	6933	3595	1428	192	38	(12)
		CDH80	14688	3	33	125	497	1400	3139	3636	3762	1614	449	26	4	(13)
(14)	Wind	WSAvg	7.0	5.9	6.8	8.0	8.2	8.5	8.6	7.4	6.4	5.9	6.5	6.2	5.7	(14)
(15)	Precipitation	PrecAvg	21.4	1.0	1.2	1.6	1.8	2.6	2.7	2.0	2.5	3.5	2.2	1.4	0.7	(15)
(16)		PrecMax	38.8	2.9	3.2	4.3	5.6	7.2	6.8	9.6	10.7	19.1	7.2	5.1	3.2	(16)
(17)		PrecMin	13.7	0.0	0.0	0.0	0.1	0.3	0.0	0.1	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	6.3	0.9	0.9	1.1	1.4	1.9	1.8	2.3	2.6	4.8	2.0	1.6	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	79.0	84.3	88.0	95.0	99.1	101.6	100.1	100.4	99.0	91.0	83.5	79.1		(19)
(20)		MCWB	54.8	57.1	59.3	61.4	67.0	69.8	69.4	69.8	67.5	67.7	62.5	55.8		(20)
(21)		DB	73.0	79.0	83.7	90.1	95.0	99.0	98.9	99.3	93.4	88.1	79.3	72.9		(21)
(22)		MCWB	54.2	56.6	58.9	62.1	68.0	69.7	69.8	70.2	69.3	67.5	61.8	56.0		(22)
(23)		DB	68.4	74.6	80.7	85.7	91.0	96.7	96.9	97.2	91.0	84.4	75.2	69.7		(23)
(24)		MCWB	53.2	55.6	59.7	62.4	67.5	69.8	69.7	70.2	68.3	66.7	60.0	55.1		(24)
(25)		DB	63.8	70.3	75.3	81.8	88.2	93.0	94.5	94.9	89.5	81.8	72.3	64.0		(25)
(26)		MCWB	51.5	55.4	58.6	62.2	67.3	69.8	70.0	70.0	68.4	65.5	59.1	54.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	64.1	65.3	67.4	70.6	74.9	76.5	75.9	75.5	75.2	74.0	69.7	64.9		(27)
(28)		MCDB	67.6	70.5	74.5	79.7	85.5	88.3	87.1	85.8	84.7	81.7	76.2	68.2		(28)
(29)		WB	60.0	63.2	65.9	69.3	73.2	75.2	74.7	74.5	73.8	71.8	67.0	62.7		(29)
(30)		MCDB	66.1	68.5	73.5	78.1	83.0	86.0	85.2	85.1	82.1	79.8	72.7	66.9		(30)
(31)		WB	56.6	61.2	64.3	68.1	72.1	74.1	73.8	73.6	72.8	70.4	64.6	59.1		(31)
(32)		MCDB	64.4	68.1	71.6	76.9	81.9	84.7	84.0	84.5	81.2	78.9	69.7	64.5		(32)
(33)		WB	52.7	58.3	63.0	66.8	71.1	73.0	73.0	73.0	71.9	69.2	62.7	55.3		(33)
(34)		MCDB	61.8	67.0	70.2	75.6	81.2	83.9	83.6	84.3	80.7	77.9	68.9	62.9		(34)
(35)	Mean Daily Temperature Range	MDBR	27.2	27.4	25.9	25.8	22.4	21.4	21.0	21.5	21.6	24.7	26.8	26.2		(35)
(36)		5% DB	36.2	35.5	31.2	30.8	27.3	25.3	23.9	24.2	25.9	25.9	30.8	35.2		(36)
(37)		MCWBR	19.8	18.9	14.3	13.8	10.1	6.8	6.8	6.7	8.9	11.0	16.3	19.5		(37)
(38)		5% WB	26.0	24.0	21.3	21.4	20.3	19.8	19.9	20.1	18.0	18.9	22.1	24.0		(38)
(39)		MCWBR	17.6	16.7	11.8	11.6	8.4	6.6	6.2	6.2	6.5	9.3	14.6	17.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.292	0.314	0.348	0.370	0.381	0.393	0.378	0.365	0.330	0.305	0.288		(40)
(41)		taud	2.574	2.564	2.509	2.427	2.423	2.437	2.450	2.491	2.457	2.549	2.548	2.568		(41)
(42)		Ebn at Noon	304	311	308	299	291	286	282	286	287	293	293	297		(42)
(43)		Edh at Noon	26	29	33	37	37	36	36	34	34	29	27	25		(43)
(44)	All-Sky Solar Radiation	RadAvg	1053	1289	1637	1954	2037	2245	2197	2040	1685	1383	1100	950		(44)
(45)		RadStd	115	170	134	152	189	163	163	110	130	198	143	120		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page